

Vectors and Vector-Valued Functions

ANSWER KEY



Vector operations in 3D

- 1 Let $\mathbf{a} = \langle 2, -1, 3 \rangle$ and $\mathbf{b} = \langle 1, 4, -2 \rangle$. Find:

a $\mathbf{a} + \mathbf{b} = \langle 3, 3, 1 \rangle$ **b** $\mathbf{a} \cdot \mathbf{b} = 2 - 4 - 6 = -8$ **c** $|\mathbf{a}| = \sqrt{14}$

- 2 Find the cross product $\mathbf{a} \times \mathbf{b}$ for $\mathbf{a} = \langle 1, 0, 2 \rangle$ and $\mathbf{b} = \langle 3, 1, -1 \rangle$.

$\mathbf{a} \times \mathbf{b} = \langle (0)(-1) - (2)(1), (2)(3) - (1)(-1), (1)(1) - (0)(3) \rangle = \langle -2, 7, 1 \rangle$.

Vector-valued functions: velocity and acceleration

- 3 A particle's position is $\mathbf{r}(t) = \langle t^2, 3t, t^3 \rangle$.

a Find the velocity vector $\mathbf{v}(t)$. $\mathbf{v}(t) = \langle 2t, 3, 3t^2 \rangle$.

b Find the acceleration vector $\mathbf{a}(t)$. $\mathbf{a}(t) = \langle 2, 0, 6t \rangle$.

c Find the speed $|\mathbf{v}(1)|$. $\mathbf{v}(1) = \langle 2, 3, 3 \rangle$, speed $= \sqrt{4 + 9 + 9} = \sqrt{22}$.

Unit tangent vectors and arc length

- 4 For $\mathbf{r}(t) = \langle 3\cos t, 3\sin t, 4t \rangle$, find $\mathbf{r}'(t)$ and the speed $|\mathbf{r}'(t)|$.

$\mathbf{r}'(t) = \langle -3\sin t, 3\cos t, 4 \rangle$. $|\mathbf{r}'(t)| = \sqrt{9\sin^2 t + 9\cos^2 t + 16} = \sqrt{9 + 16} = 5$ (constant speed).

- 5 Find the arc length of $\mathbf{r}(t) = \langle 3\cos t, 3\sin t, 4t \rangle$ for $0 \leq t \leq 2\pi$, using the constant speed found above.

$L = \int_0^{2\pi} |\mathbf{r}'(t)| dt = \int_0^{2\pi} 5 dt = 10\pi$.

Applications of the cross product

- 6 Find a vector orthogonal to both $\mathbf{u} = \langle 1, 2, 0 \rangle$ and $\mathbf{v} = \langle 0, 1, 3 \rangle$, using the cross product.

$\mathbf{u} \times \mathbf{v} = \langle (2)(3) - (0)(1), (0)(0) - (1)(3), (1)(1) - (2)(0) \rangle = \langle 6, -3, 1 \rangle$.

- 7 Find the area of the parallelogram spanned by $\mathbf{u} = \langle 2, 0, 0 \rangle$ and $\mathbf{v} = \langle 1, 3, 0 \rangle$, using $|\mathbf{u} \times \mathbf{v}|$.

$\mathbf{u} \times \mathbf{v} = \langle 0, 0, 6 \rangle$. Area $= |\langle 0, 0, 6 \rangle| = 6$.

Lines and planes in space

- 8 Write parametric equations for the line through $(1, 2, -1)$ in the direction of $\mathbf{v} = \langle 3, -1, 2 \rangle$.

$x = 1 + 3t, y = 2 - t, z = -1 + 2t$.

- 9 Find the equation of the plane through $(1, 0, 2)$ with normal vector $\mathbf{n} = \langle 2, -3, 1 \rangle$.
 $2(x - 1) - 3(y - 0) + 1(z - 2) = 0$, i.e. $2x - 3y + z = 4$.
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Motion along a curve

- 10 A particle's position is $\mathbf{r}(t) = \langle e^t, e^{-t}, \sqrt{2}t \rangle$. Find $\mathbf{v}(t)$ and show that the speed $|\mathbf{v}(t)|$ is minimized at $t = 0$.
 $\mathbf{v}(t) = \langle e^t, -e^{-t}, \sqrt{2} \rangle$. $|\mathbf{v}(t)|^2 = e^{2t} + e^{-2t} + 2$. By AM-GM, $e^{2t} + e^{-2t} \geq 2$ with equality at $t = 0$, so $|\mathbf{v}(t)|^2 \geq 4$, minimized (equal to 4) at $t = 0$.